

Chapter 5

Enthalpy

1. Which one of the following conditions would always result in an increase in the internal energy of a system D?

- A) The system loses heat and does work on the surroundings.
- B) The system gains heat and does work on the surroundings.
- C) The system loses heat and has work done on it by the surroundings.
- D) The system gains heat and has work done on it by the surroundings.
- E) None of the above is correct.

2. Which one of the following statements is true C?

A) Enthalpy is an intensive property. $\propto \text{Mass}$

B) The enthalpy change for a reaction is ~~independent~~ of the state of the reactants and products.

C) Enthalpy is a state function. ✓

D) H is the value of q measured under conditions of constant ~~volume~~. Pressure

$$\Delta H = \Delta(E + PV) = \Delta E + P\Delta V = (\overset{\text{First law}}{q + w}) - w$$

E) The enthalpy change of a reaction is the ~~reciprocal~~ $\frac{1}{\Delta H}$ of the ΔH of the reverse reaction.

3. The reaction



is _____, and therefore heat is _____ **C** _____ by the reaction.

- A) endothermic, released
- B) endothermic, absorbed
- C) exothermic, released
- D) exothermic, absorbed
- E) thermoneutral, neither released nor absorbed

4. Under what condition(s) is the enthalpy change of a process equal to the amount of heat transferred into or out of the system

B ?

(a) temperature is constant

(b) pressure is constant

(c) volume is constant

$$\begin{aligned}\Delta H &= \Delta E + P\Delta V \\ &= q + w - w = q\end{aligned}$$

A) a only

B) b only

C) c only

D) a and b

E) b and c

5. The British thermal unit (Btu) is commonly used in engineering applications. A Btu is defined as the amount of heat required to raise the temperature of 1 lb of water by 1 °F. There are _____^C joules in one Btu. 1 lb = 453.59 g; °C = (5/9)(°F - 32°); specific heat of H₂O(l) = 4.184 J/g·K.

$$1 \text{ lb} \cdot \frac{453.59 \text{ g}}{1 \text{ lb}} \cdot 4.184 \text{ J/g}\cdot\text{K} \cdot 1.8 \frac{^\circ\text{F}}{^\circ\text{C}} =$$

- A) 3415
- B) 60.29
- C) 1054
- D) 5.120×10^{-3}
- E) Additional information is needed to complete the calculation.

6. A 22.44 g sample of iron absorbs 180.8 J of heat, upon which the temperature of the sample increases from 21.1 °C to 39.0 °C. What is the specific heat of iron B ?

- A) 0.140
- B) 0.450
- C) 0.820
- D) 0.840
- E) 0.900

7. Of the following, ΔH_f° is not zero for D.

A) $\text{O}_2 (\text{g})$

B) $\text{C} (\text{s, graphite})$

C) $\text{N}_2 (\text{g})$

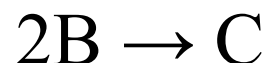
D) $\text{F}_2 (\text{s})$

E) $\text{Cl}_2 (\text{g})$

8. Consider the following two reactions:



Determine the enthalpy change for the process: A



A) -478.8 kJ/mol

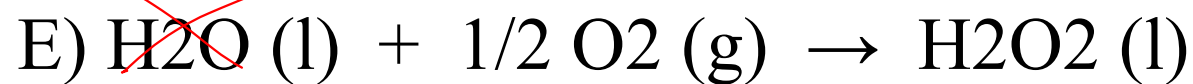
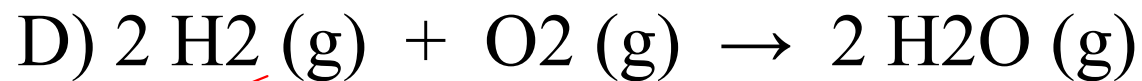
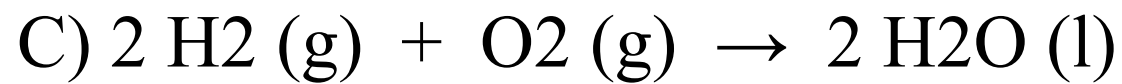
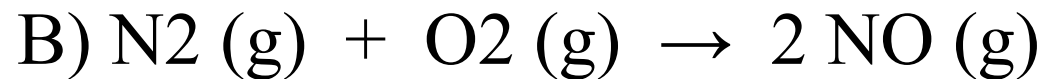
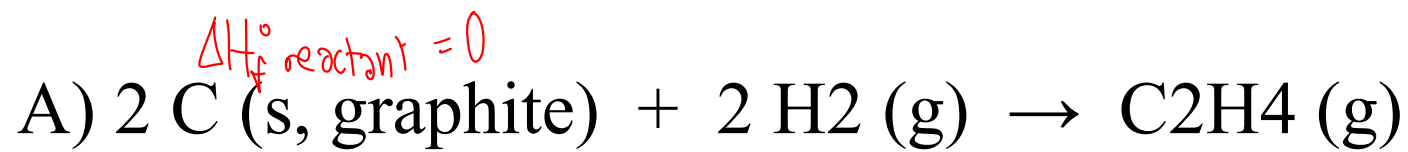
B) -434.6 kJ/mol

C) 434.6 kJ/mol

D) 478.8 kJ/mol

E) More information is needed to solve the problem.

9. For which one of the following reactions is the value of $\Delta H^\circ_{\text{rxn}}$ equal to ΔH°_f for the product A?



4.500000
1 atm 4.500000